ASSIGNMENT REPORT– 8

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For the Subject

Pattern Recognition and Machine Learning

(AI5707)

DEPARTMENT OF COMPUTER SCIENCE

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Classification Using GMM, LDA, KNN, SVM and DTW

1. Objective

The objective of this assignment is to implement and compare the different classification using four dataset synthetic data, MNIST ,TI Digits and image features.

The classifiers used are Gaussian mixture model, Linear discrimination analysis, K- nearest neighbors , Support vector machine, Dynamic time wrapping .Principle component analysis is used to reduce the number of features and show the classification accuracy .

2. Dataset Used

2.1Synthetic Dataset

Synthetic dataset is loaded from data.pkl file that contain multiple input features x and class labels y . Each samples contain 16 features. We load features and class labels from pickle file then divide them into training , validation and testing sets.

2.2MNIST Dataset

MNIST dataset contains grayscale images of handwritten digits . The assigned digits are 1,3,5. Each images is resized to 28x28 pixels and converted into 784 numerical pixel features.

2.3TI Digits Dataset

TI Digits dataset contain speech recordings of spoken digits. Our team assigned digit 4,8,9 . the data are loaded from .npz files as sequence. Convert the recording into fixed length features for LDA, KNN, GMM and SVM , DTW.

2.4Image Features Dataset

Image Features dataset contains numerical features extracted from images belongs to 5 classes . Each contain 36 features and class label . Read train, validation, test data from CSV file .

3.Methodology

1.Load the dataset and divide data into training , validation and testing.

2.Preprocess and standardize features of dataset .

3.Implement the classifiers LDA, KNN, GMM , and SVM .

- LDA : The mean and covariance of each class using training data are calculated . we use the values to calculate scores and predict class of each test sample.
- KNN: Distance between each test sample and training samples are calculated. Select k nearest sample and predict the class based on high value.
- GMM: Model parameter and calculated likelihood of each test to predict its class.
- SVM: Train SVM to separate different classes using training features . Then used trained model to predict class label of test samples

4.Evaluate each classifiers accuracy using the original features.

5. DTW is implemented for the TI digits .

6.After finding the accuracy of all classifiers now apply PCA .

7.Apply PCA to reduce the feature dimensions then again evaluate the classifiers.

8.Compare accuracies , confusion matrices, and final result .

4.Result

4.1Synthetic Dataset

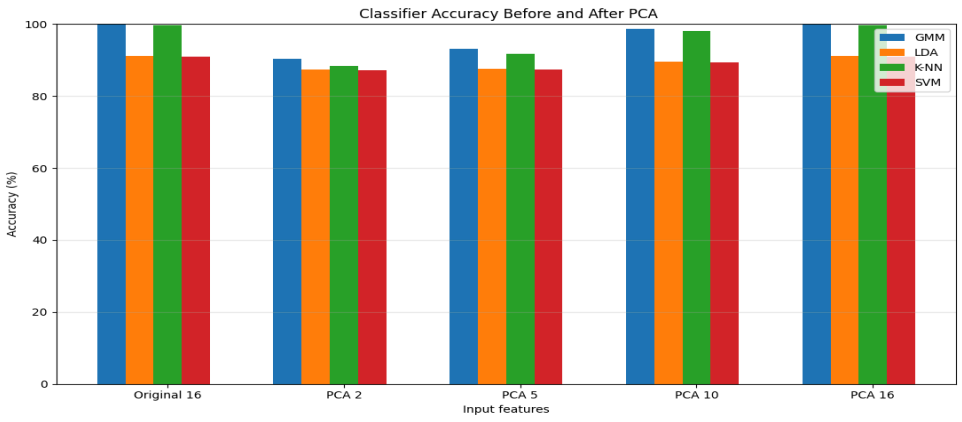
Original dataset accuracy

```
validation accuracy
GMM accuracy: 100.0 %
LDA accuracy: 91.2 %
K-NN accuracy: 99.6 %
SVM accuracy: 90.9 %
test accuracy
GMM accuracy: 99.97 %
LDA accuracy: 91.57 %
K-NN accuracy: 99.5 %
SVM accuracy: 90.03 %
```

Gmm performs best when compared to the rest of the classifiers with validation accuracy 100% and test accuracy 99.97% .

pca features accuracy:(K=2,5,10,16)

Features	GMM	LDA	K-NN	SVM
Original 16	100.00%	91.20%	99.60%	90.90%
PCA 2	90.30%	87.40%	88.30%	87.27%
PCA 5	93.13%	87.53%	91.67%	87.30%
PCA 10	98.67%	89.63%	98.07%	89.37%
PCA 16	100.00%	91.20%	99.60%	90.90%



GMM performs best even when the features are reduced using pca k=2,5,10,16 . GMM has the highest accuracy value when feature value(k=16).

4.2.MNIST dataset

Original dataset accuracy

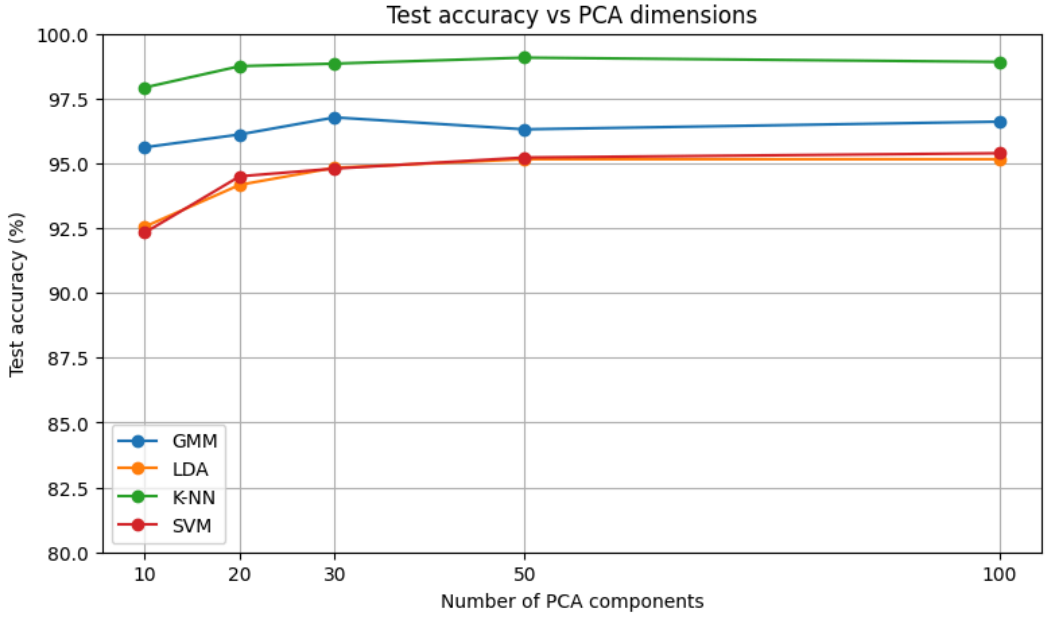
```
========== DEV ACCURACY ==========
GMM accuracy: 81.49 %
LDA accuracy: 87.45 %
K-NN accuracy: 99.18 %
SVM accuracy: 94.23 %

========== TEST ACCURACY ==========
GMM accuracy: 82.94 %
LDA accuracy: 87.69 %
K-NN accuracy: 98.85 %
SVM accuracy: 95.16 %
```

The k-nn classifier has the highest test accuracy(98.85%) and validation accuracy(99.18%). The GMM classifier has the lowest accuracy.

pca features accuracy:(K=10,20,30,50,100)

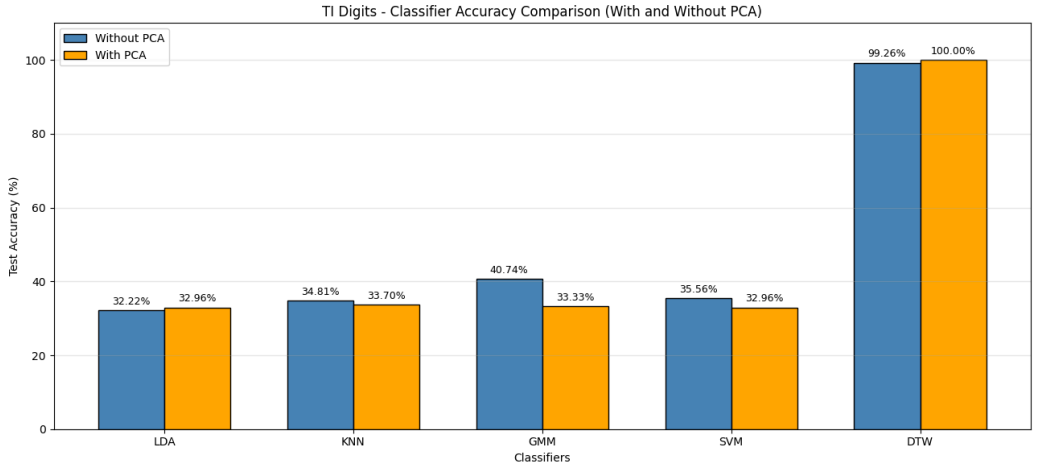
Classifier	GMM	K-NN	LDA	SVM
PCA Components				
10	95.620678	97.925584	92.558446	92.327955
20	96.114587	98.748765	94.171880	94.501152
30	96.773131	98.847547	94.830425	94.797498
50	96.312150	99.078038	95.159697	95.225552
100	96.608495	98.913401	95.159697	95.390188



K-NN performs best even when the features are reduced using pca (k=10,20,30,50,100). K-NN has the highest accuracy at K=50.

4.3 TI Digits Datasets

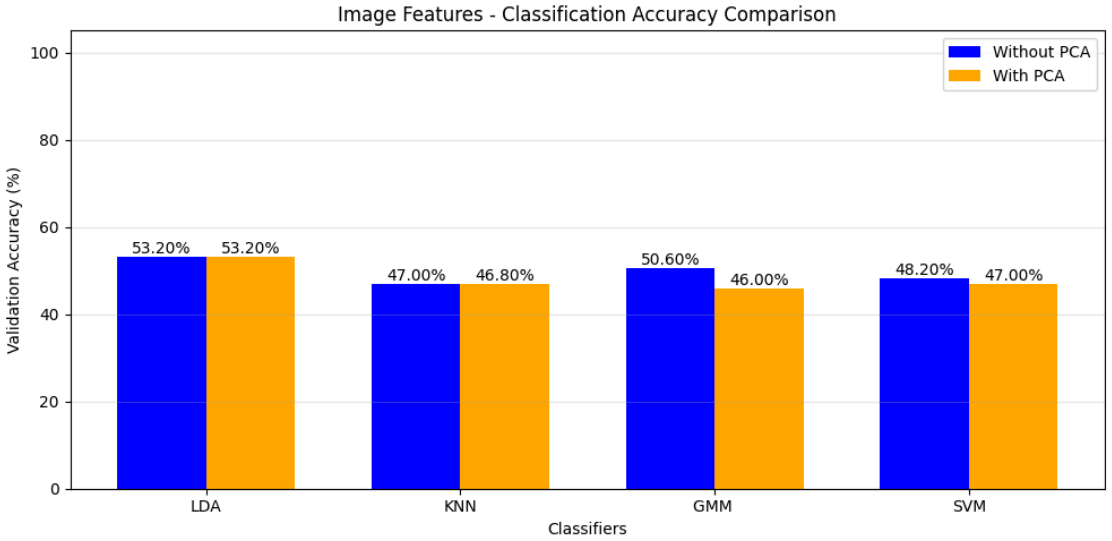
```
TI DIGITS - FINAL CLASSIFIER COMPARISON
------------------------------------------------------------
Classifier     Without PCA         With PCA
------------------------------------------------------------
LDA            32.22               32.96
KNN            34.81               33.70
GMM            40.74               33.33
SVM            35.56               32.96
DTW            99.26               100.00
------------------------------------------------------------
```



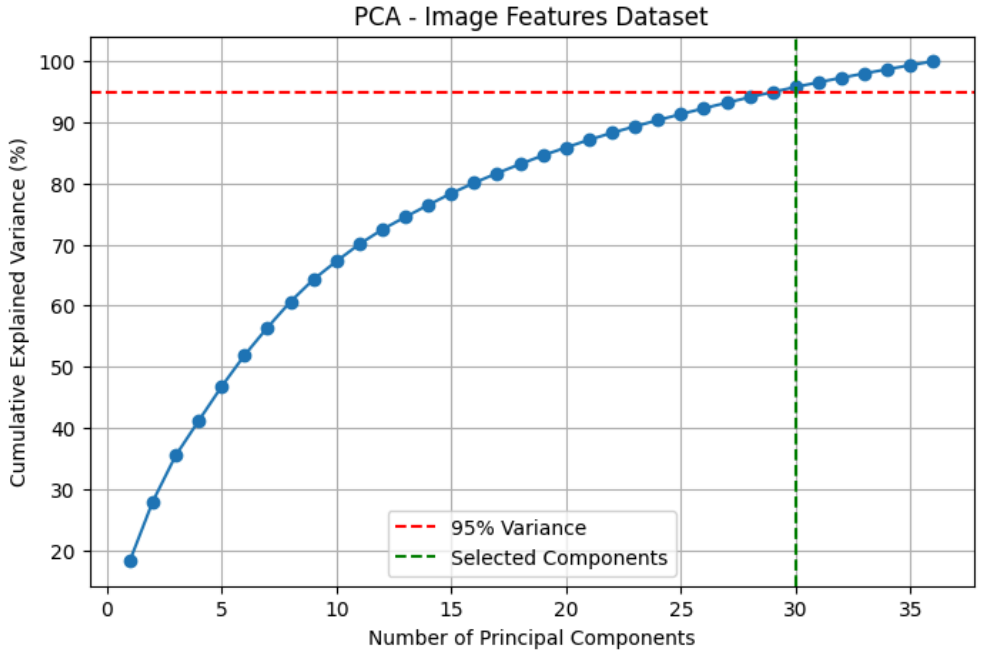
Observation: DWT performed better than the other classifiers . DTW with PCA results achieved 100% accuracy.

4.4 Image Features Dataset

```
FINAL TEST RESULTS - IMAGE FEATURES
------------------------------------------------
LDA Original        : 53.20%
KNN Original        : 45.00%
GMM Original        : 52.40%
SVM Original        : 49.40%
LDA PCA             : 53.20%
KNN PCA             : 41.60%
GMM PCA             : 44.40%
SVM PCA             : 49.00%
```



Observation: LDA achieved the highest accuracy of 53.20%.



5. Inference

- In this assignment we implemented classification and tested using four different datasets.
- It show that each classifiers performs differently by applying different dataset.
- Synthetic data GMM get highest accuracy of 100%. MNIST dataset KNN get highest accuracy of 99%. TI digits dataset DTW V get highest accuracy of 100%. And Image features dataset LDA get highest accuracy of 53.20%.
- PCA is applied to all four dataset to reduce the input feature while retaining the important information
- And the classifiers were tested again using these reduced features.